# LAB 7 REPORT

# Fuzzing for Input Validation Bugs

**Name:** PRITHVI RAJ PANDEY
**SRN:** PES1UG25CS833
**Section:** C

## Objective

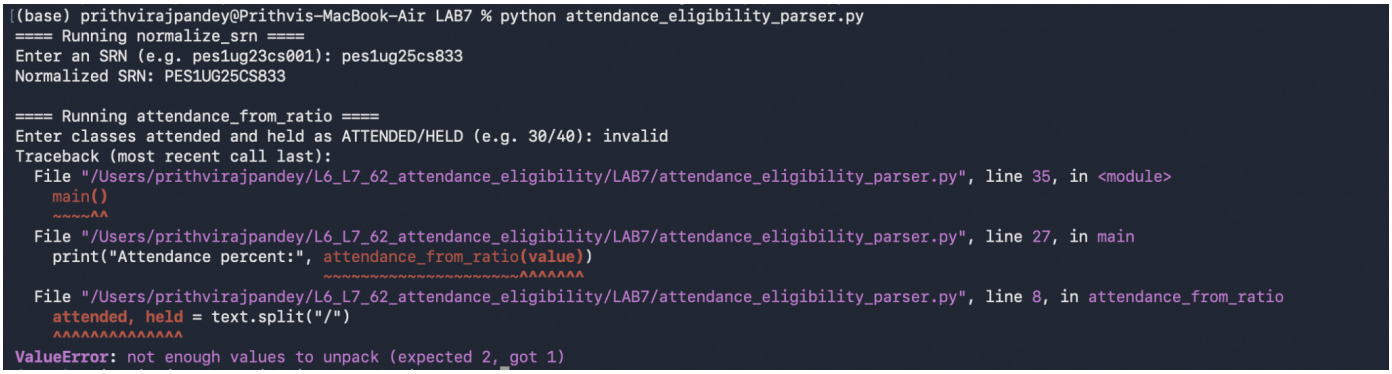
To explore fuzz testing as a security and quality assurance technique by identifying and fixing input validation bugs in a Python program using the Hypothesis fuzzing library.

## Summary of Work

The three functions in the attendance eligibility parser were tested with automatically generated unexpected inputs. The initial implementation crashed for inputs such as None, malformed attendance ratios, non-numeric values, and empty registers. The functions were then hardened with input validation so that invalid inputs are handled safely without crashing.

## SS1 — Running the buggy module

The buggy application is run with malformed attendance-ratio input. The program terminates with a ValueError, demonstrating an input validation bug.




## SS2 — Hypothesis tests failing on the buggy module

Property-based fuzzing generates unexpected inputs and exposes crashes in all three functions.

```
(base) prithvirajpandey@Prithvis-MacBook-Air LAB7 % pytest test_attendance_eligibility_parser.py -v
============================= test session starts ==============================
platform darwin -- Python 3.14.6, pytest-9.2.0, pluggy-1.6.0 -- /opt/miniconda3/bin/python3.14
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /Users/prithvirajpandey/L6_L7_62_attendance_eligibility/LAB7
plugins: anyio-4.12.1, hypothesis-6.170.15
collected 3 items

test_attendance_eligibility_parser.py::test_normalize_srn_no_crash FAILED [ 33%]
test_attendance_eligibility_parser.py::test_attendance_from_ratio_no_crash FAILED [ 66%]
test_attendance_eligibility_parser.py::test_attendance_from_register_no_crash FAILED [100%]

=================================== FAILURES ===================================
_____________________________ test_normalize_srn_no_crash _____________________________

    @given(st.text() | st.none())
>   def test_normalize_srn_no_crash(s):
                    ^^^

test_attendance_eligibility_parser.py:11:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
test_attendance_eligibility_parser.py:12: in test_normalize_srn_no_crash
    normalize_srn(s)
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

srn = None

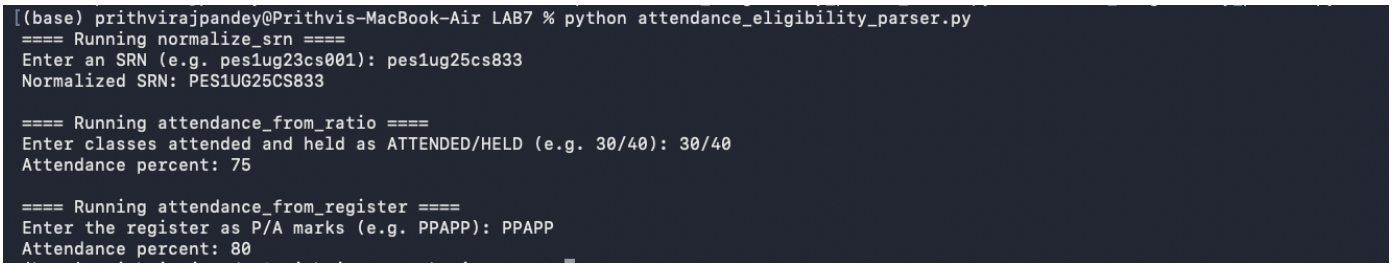
    def normalize_srn(srn):
        """Trims and upper-cases a student registration number."""
>       return srn.strip().replace(" ", "").upper()
                ^^^^^^^^

E       AttributeError: 'NoneType' object has no attribute 'strip'
E       Falsifying test case: test_normalize_srn_no_crash(
E           s=None,
E       )

attendance_eligibility_parser.py:3: AttributeError
_____________________ test_attendance_from_ratio_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/prithvirajpandey/L6_L7_62_attendance_eligibility/LAB7/test_attendance_eligibility_parser.py", line 16, in test_attendance_from_ratio_no_crash
  |     def test_attendance_from_ratio_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/prithvirajpandey/L6_L7_62_attendance_eligibility/LAB7/test_attendance_eligibility_parser.py", line 17, in test_attendance_from_ratio_no_crash
    |     attendance_from_ratio(s)
    |     ~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "/Users/prithvirajpandey/L6_L7_62_attendance_eligibility/LAB7/attendance_eligibility_parser.py", line 9, in attendance_from_ratio
    |     return int(attended.strip()) * 100 // int(held.strip())
    |                ~~~^^^^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Falling test case: test_attendance_from_ratio_no_crash(
    |   s='/',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/prithvirajpandey/L6_L7_62_attendance_eligibility/LAB7/test_attendance_eligibility_parser.py", line 17, in test_attendance_from_ratio_no_crash
    |     attendance_from_ratio(s)
    |     ~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "/Users/prithvirajpandey/L6_L7_62_attendance_eligibility/LAB7/attendance_eligibility_parser.py", line 8, in attendance_from_ratio
    |     attended, held = text.split("/")
    |     ^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Falling test case: test_attendance_from_ratio_no_crash(
    |   s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/prithvirajpandey/L6_L7_62_attendance_eligibility/LAB7/test_attendance_eligibility_parser.py", line 17, in test_attendance_from_ratio_no_crash
    |     attendance_from_ratio(s)
    |     ~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "/Users/prithvirajpandey/L6_L7_62_attendance_eligibility/LAB7/attendance_eligibility_parser.py", line 8, in attendance_from_ratio
    |     attended, held = text.split("/")
    |                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Falling test case: test_attendance_from_ratio_no_crash(
    |   s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_attendance_from_register_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/prithvirajpandey/L6_L7_62_attendance_eligibility/LAB7/test_attendance_eligibility_parser.py", line 21, in test_attendance_from_register_no_crash
  |     def test_attendance_from_register_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/prithvirajpandey/L6_L7_62_attendance_eligibility/LAB7/test_attendance_eligibility_parser.py", line 22, in test_attendance_from_register_no_crash
```

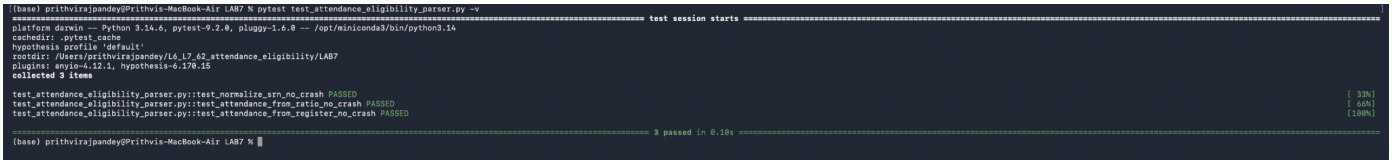
## SS3 — Running the fixed module

The fixed application is executed with valid sample inputs. It successfully normalizes the SRN and calculates attendance percentages.




## SS4 — Hypothesis tests passing on the fixed module

After the fixes, all three Hypothesis tests pass, confirming that the functions no longer crash for the generated text and None inputs.



## Reflection

**1. What bugs were discovered through fuzz testing?**

Fuzz testing exposed several unhandled edge cases. normalize_srn crashed when given None because string methods were called on a non-string value. attendance_from_ratio crashed for malformed ratios, missing separators, non-numeric values, and other invalid inputs. attendance_from_register could also fail when given None or an empty value.

**2. How were the bugs fixed?**

Defensive input validation was added to each function. The functions now check the type and structure of incoming data before processing it. Invalid or unexpected inputs are handled safely instead of allowing exceptions such as AttributeError, ValueError, or division-related errors to terminate the program.

**3. What did you learn from this lab?**

The lab demonstrated that normal manual testing can miss unusual inputs and edge cases. Hypothesis property-based testing automatically generates many unexpected inputs and helps find the smallest failing examples. Fuzzing is therefore useful for improving both software reliability and robustness.

## Conclusion

The attendance eligibility parser was successfully tested using Hypothesis fuzzing and improved through defensive programming. The final test run shows all three tests passing.